



Removal of diclofenac from wastewater: A comprehensive review of detection, characteristics and tertiary treatment techniques

Ingridy Alessandretti^a, Cesar Vinicius Toniciolli Rigueto^{b,*}, Mateus Torres Nazari^c,
Marieli Rosseto^b, Aline Dettmer^{a,*}

^a University of Passo Fundo (UPF), Faculty of Engineering and Architecture (FEAR), Chemical Engineering Course, Passo Fundo, RS, Brazil

^b Federal University of Santa Maria (UFSM), Postgraduate Program in Food Science and Technology (PPGCTA), Santa Maria, RS, Brazil

^c University of Passo Fundo (UPF), Faculty of Engineering and Architecture (FEAR), Postgraduate Program in Civil and Environmental Engineering (PPGEng), Passo Fundo, RS, Brazil

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ABSTRACT

The increase in the consumption of diclofenac (DCF), the incorrect disposal of this pharmaceutical and the low efficiency of removal in wastewater treatment plants result in the detection of this emerging contaminant in different environmental compartments, especially in water resources. DCF can be toxic for several organisms according to its concentration. In this sense, the present review presents the occurrence of DCF in aqueous matrices around the world, the main characteristics of this contaminant and some alternative methods for DCF removal from synthetic aqueous media and real effluents, such as adsorption, membrane separation, and advanced oxidation processes. A bibliometric and systematic approach was performed to systematize the state-of-the-art of the last 10 years, using Scopus as a database. It was concluded that the techniques presented are effective to DCF treatment, since a removal efficiency of up to 100% was found in the literature reviewed for each sole treatment. However, these techniques present limitations related to the large-scale efficiency, high costs, intermediate compounds production, and complexity of the mechanisms of action. The integration of technologies is a promising alternative to overcome these limitations, but the implementation of a coupled system must consider several factors, especially for real effluents. Finally, we show how the advances in DCF techniques removal from aqueous matrices can positively impact several Sustainable Development Goals (SDGs) of the United Nations.

1. Introduction

The global production of pharmaceutical products in 2019 was valued at 324.42 billion dollars, with an annual growth forecast of 13.74% for the period from 2020 to 2027 [1]. Pharmaceuticals are responsible for increasing life-expectancy, thus there is an increase in their consumption, which are often released and detected in surface-, waste- and groundwater, as known as emerging contaminants [2–4]. These contaminants have been detected in the environment in unchanged and metabolized forms at low concentrations (ng L^{-1} to mg L^{-1}), but their solubility, bioaccumulation, persistence, biotransformation and toxicity characteristics can cause risks to human health and the ecosystem [2,5,6].

Diclofenac (DCF) is an example of emerging contaminant which can be sodium or potassium, both classified as non-steroidal anti-

inflammatory drugs (NSAIDs), used as analgesics for application in humans, livestock, and domestic animals, in the treatment of inflammation and pain in pathologies [7]. The global consumption of DCF has been estimated at around 940 tons per year, with an average of 65% of the oral dosage of this pharmaceutical being released through the urine with its active metabolisms, that are forwarded to the conventional wastewater treatment plants (WWTPs) [8]. Other ways of these contaminants being present in superficial and wastewater is through their improper disposal as solid wastes, excretion of pharmaceutical and their metabolites through humans and veterinary routes, and by effluents from industrial and urban wastewater treatment plants (WWTPs and UWWTPs), due to the ineffectiveness of conventional treatments [9,10].

In this context, some studies have addressed the effect of diclofenac toxicity in different species, as in rat, brown trout and sea urchin, which effects resulted in an increase of mortality, damage to the tissues and

* Corresponding authors.

E-mail addresses: cesartoniciolli@acad.ufsm.br (C.V.T. Rigueto), alinedettmer@upf.br (A. Dettmer).

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biochemical functions, in addition to gastrointestinal lesions and interference in growth rate and development have also been reported [11–14].

Based on the toxicity of this compound and its presence in surface and wastewater, the European Commission has defined a document referring to environmental quality standards (EQS), which contains chronic toxicity limits for the annual average and acute toxicity for the maximum permitted concentrations, which are $0.1\text{--}0.01\ \mu\text{g L}^{-1}$ and $75\text{--}7.5\ \mu\text{g L}^{-1}$ for DCF, respectively, both for inland and coastal surface water [5,15].

In order to achieve the permitted concentrations, some tertiary techniques have been developed to remove emerging contaminants from aqueous matrices, such as adsorption [16–19], membrane separation [20–23] and advanced oxidation processes [24–27]. These techniques have been used to remove traces of various contaminants, including diclofenac.

According to Taoufik et al. [28], diclofenac is the most pharmaceutical contaminant studied by scientific community aiming its removal by different treatment techniques. In this sense, the aim of this study was to carry out a review that reports the main tertiary techniques used for the removal of diclofenac from aqueous media, as well as the main parameters of each technique, removal efficiencies, and economic and environmental aspects involved in this context.

2. Methodology

The survey of the articles was carried out by inserting specific terms in the Scopus database. The terms "TECHNIQUE" AND "REMOVAL" AND "DICLOFENAC" were inserted, and the documents with these words in the title, abstract and keywords were analyzed. The search period was limited from 2010 to 2021. The results obtained from the preliminary search were explored using the "Bibliometrix" tool of RStudio® software version 1.4.1106, to eliminate overlapping articles that did not fit the scope of the work and also to separate the articles in relation to the techniques considered most appropriate based on their content. The methodology of this study is shown in following the flowchart (Fig. 1).

From the bibliometric analysis, it can be seen that interest in the thematic of diclofenac removal is growing, with an annual growth rate of 4.75%, and peak of publication in the year of 2020 (Fig. 2a). The

Fig. 2b presents that China is the country that most published articles focused on diclofenac removal techniques, followed by India, Spain, Italy and Brazil.

In the WordCloud with the 50 most cited words in the keywords of the analyzed articles (Fig. 2c), it is clear that the main techniques used to remove diclofenac in the studies are adsorption, membrane separation and advanced oxidation processes. Therefore, these techniques were explored in the topics of this review.

3. Diclofenac: characteristics and detection in aqueous matrices

3.1. Characteristics

Diclofenac is used to relieve pain and reduce inflammation, which can be applied to the skin or orally, being one of the most used NSAIDs in the world, and its market share is close to ibuprofen, mefenamic acid, naproxen [10]. This pharmaceutical can be formulated from two salts, sodium and potassium, the difference between the two is the absorption time, with sodium DCF producing a slower absorption, with release in the duodenum, used for pain chronic, while potassium DCF has a quick release, with absorption in the stomach and thus used for acute pain [29]. Table 1 shows some important physicochemical properties of diclofenac sodium and potassium.

Octanol-water partition coefficient ($\log K_{ow}$ or $\log P$) is used to measure the hydrophilicity/lipophilicity of a compound, being useful in determining the bioaccumulation potential and toxicity of a given compound in relation to the environment or living organism [30]. $\log K_{ow}$ values range from -3 (extremely hydrophilic) to $+10$ (extremely hydrophobic) [30], DCF has a $\log K_{ow}$ of around 4.5, which according to the US EPA [31], $\log K_{ow}$ values higher than 4 are considered hostile to aquatic life. Therefore, substances with high values of $\log K_{ow}$, tend to be absorbed by the soil and on the contrary tend to remain in the water; however, these compounds can be remobilized, depending on the binding force [32]. Another parameter that confirms the lipophilicity of diclofenac is a dipole moment value of 2.51 D [23,33].

The dissociation constant (pK_a) is used to compare the relative acidity or basicity of weakly ionizing compounds in aqueous solvent solutions or the miscibility between solute and solvent. DCF has a pK_a of 4.15, which indicates that at $pH < 4.15$, the predominant species is

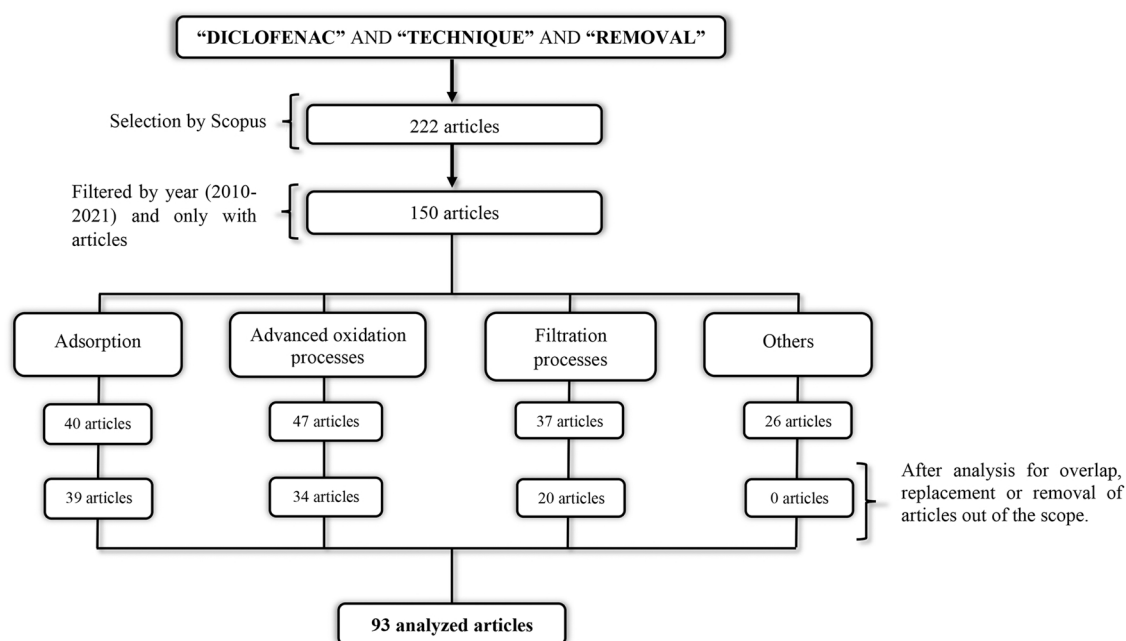
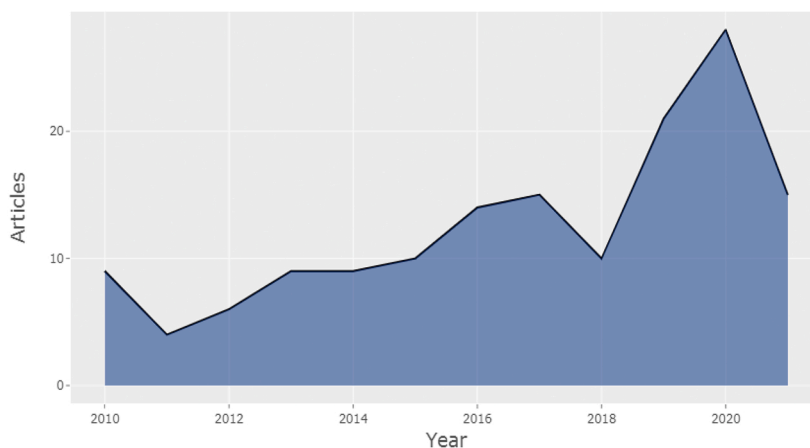
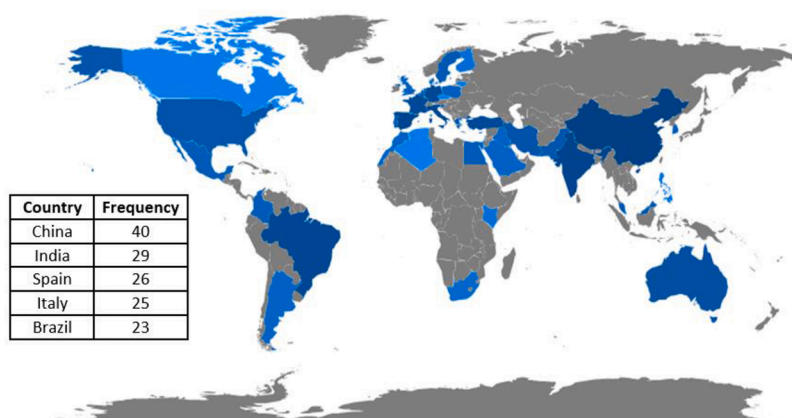


Fig. 1. Conditions established for the selection and analysis of articles within the scope of this study.

Annual Scientific Production

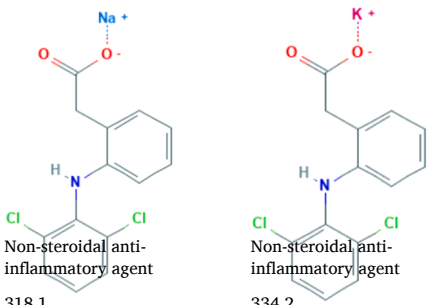


Country Scientific Production

[illegible]

3

Table 1
Diclofenac physicochemical properties.

CAS Number	Diclofenac sodium 15307–79-6	Diclofenac potassium 15307–81-0
Formula	$C_{14}H_{10}Cl_2NNaO_2$	$C_{14}H_{10}Cl_2KNO_2$
Structure		
Therapeutic Class	Non-steroidal anti-inflammatory agent	
Molecular Mass (g mol ⁻¹)	318.1	334.2
log K _{ow}	4.51	4.28
pKa	4.15	4.10
Water solubility (mg.mL ⁻¹)*	2.37	2.34

Source: Extracted from [33–36].

protonated (neutral), while at pH > 4.15, the majority form is deprotonated (monoanionic). pH can affect the effectiveness of some DCF removal techniques [37,38].

Diclofenac has low water solubility, so sodium and potassium salts are used to improve its solubility and thus the absorption of the compound, which makes it more soluble and present in aqueous matrices [10,38]. The solubility of diclofenac is affected by the pH of the solution due to the pKa of the compound, so at pH lower than the pKa (4.15), the DCF has a neutral form and then has its solubility decreased, which causes DCF precipitation in these conditions [39].

3.2. Diclofenac detection in aqueous matrices

Table 2 presents studies that reported the presence of diclofenac in several aqueous matrices, addressing the nature and local of sample, and the DCF detected concentration.

It is observed that the countries that presented the most studies regarding the detection of diclofenac in various aqueous matrices in the period 2019–2021 were China, Poland, Portugal, United States and Spain (Table 2). China and Spain are in the list of countries that have published the most about DCF removal techniques from aqueous media (Fig. 2b).

Regarding the detection taking into account the continents, there is an emphasis on countries on the European continent, as the DCF was inserted into the Environmental Quality Standards (EQS) of Europe, which reports that the maximum concentration of DCF in fresh water and marine water are 0.1 and 0.01 µg.L⁻¹, respectively. In addition, in 2013 it was included in the list of “EU Water Framework Directive”, with the aim of sampling, monitoring and adopting measures to reduce risks [10]. On other continents, such as Asia, the lower frequency of detection studies is due to the lack of measures for monitoring DCF in aqueous matrices [10].

The conventional treatment of wastewater consists of the pre-treatment steps (grilling, desanding and sieving), primary treatments (sedimentation, coagulation and flocculation), secondary treatments (aerobic and anaerobic processes, such as activated sludge, UASB reactor and stabilization and anaerobic lagoons), which aim to remove total suspended solids (TSS), color, turbidity and organic matter (BOD and COD) [5]. According to Matamoros et al. [63] the removal of TSS by conventional processes varies according to the techniques used in each

Table 2
Detection of diclofenac reported from 2019 to 2021 in several aqueous matrices.

Nº	Nature of sample	Concentration (ng L ⁻¹) 1)	Country	Reference
1.	Surface water	60.70	Portugal	[40]
2.	Surface water	1130.00	India	[41]
3.	Wastewater ²	3160.00	Latvia	[42]
4.	Wastewater ¹	570.00	Latvia	[42]
5.	Wastewater ²	3700.00	Germany	[43]
6.	Wastewater ¹	1800.00	Germany	[43]
7.	Wastewater ¹	1142.00	United States	[44]
8.	Wastewater ²	353.00	Spain	[45]
9.	Wastewater ¹	572.00	Spain	[45]
10.	Surface water	169.00	Spain	[45]
11.	Wastewater ²	31.45	China	[3]
12.	Wastewater ²	45.30	China	[3]
13.	Surface water	200.00	Nigeria	[4]
14.	Groundwater	42.00	Nigeria	[4]
15.	Drinking water	< 1.00	Nigeria	[4]
16.	Surface water	10005.00	South Africa	[46]
17.	Wastewater ²	328.50	Australia	[47]
18.	Wastewater ¹	202.50	Australia	[47]
19.	Wastewater ²	467.00	United Kingdom	[48]
20.	Wastewater ²	143.00	United States	[48]
21.	Wastewater ²	338.00	Mexico	[48]
22.	Surface water	1390.00	Saudi Arabia	[49]
23.	Seawater	79.80	Brazil	[50]
24.	Seawater	346.00	Portugal	[51]
25.	Groundwater	606.10	Poland	[52]
26.	Wastewater ¹	3614.00	Egypt	[53]
27.	Surface water	77.00	Egypt	[53]
28.	Wastewater ¹	9500.00	Iran	[54]
29.	Drinking water	550.00	Germany	[55]
30.	Wastewater ²	4450.00	Germany	[55]
31.	Wastewater ¹	4280.00	Germany	[55]
32.	Wastewater ²	3139.00	Poland	[56]
33.	Wastewater ¹	4139.00	Poland	[56]
34.	Wastewater ¹	220.00	China	[57]
35.	Surface water	200.00	China	[57]
36.	Wastewater ¹	810.00	Spain	[58]
37.	Wastewater ¹	1220.00	Spain	[59]
38.	Wastewater ²	570.00	United States	[60]
39.	Wastewater ¹	63.00	United States	[60]
40.	Wastewater ¹	246.30	Hong Kong	[61]
41.	Wastewater ¹	154.06	India	[61]
42.	Wastewater ¹	1.40	Philippines	[61]
43.	Wastewater ¹	13.70	United States	[61]
44.	Wastewater ¹	57.20	Switzerland	[61]
45.	Wastewater ¹	6270.60	Sweden	[61]
46.	Surface water	388.00	Portugal	[62]
47.	Surface water	1856.00	Portugal	[62]

¹effluent or ²influent.

primary and secondary process, about 94% of the TSS being removed. Therefore, the DCF dissolved in the aqueous medium not previously removed can be treated using tertiary techniques [64].

As already mentioned, conventional techniques are not effective for complete removal of organic contaminants. Therefore, adsorption, membrane separation and advanced oxidation processes can be used as tertiary techniques for the treatment of wastewater containing pharmaceuticals, such as diclofenac [5,15,80].

4. Tertiary techniques for diclofenac removal

Due to the low concentrations detected in aquatic environments and the complex chemical nature of diclofenac, conventional coagulation and sedimentation techniques generally reduce less than 25% of the initial pharmaceutical concentration present in aqueous media [65]. In this sense, it is necessary to improve the removal efficiency of DCF and other emerging contaminants. Some tertiary techniques have been studied to achieve this aim, such as adsorption, advanced oxidation processes and membrane separation.

4.1. Adsorption

The adsorption technique has been widely used in studies for the removal of a range of contaminants due to its simplicity, high removal rate, ease of operation and implementation, low cost, and absence of sludge formation [66]. The main contaminants reported in adsorption studies are dyes [67], toxic metals [37], oily contaminants [68], and pharmaceuticals and personal care products [69,70].

Among pharmaceutical products, diclofenac has to get prominence due to its toxicity and bioaccumulation observed in aquatic organisms, even in low concentrations. Besides, studies have reported that the metabolites of diclofenac can be generated from biotransformation by living organisms or by exposure to sunlight, and some of them are even more toxic than pure diclofenac [71]. In this context, the removal of diclofenac in an aqueous medium by adsorption is a promising alternative, as it does not transfer the emerging contaminants into other transformation products or generate any genotoxic by-products [6,72].

To use the adsorption technique for wastewater treatment, the steps for choosing, developing, and characterizing the adsorbent materials are

fundamental points since the adsorbent materials must have desirable characteristics, such as low cost, wide availability, chemical, mechanical and thermal stability, good adsorption capacity, fast adsorption kinetics, high selectivity, good physicochemical and texture characteristics, in addition to the possibility of reuse and/or regeneration [73].

Activated carbon is a commercial material commonly used for the adsorption of contaminants. However, the cost of activated carbon often restricts its viability at the economic level. Therefore, the search for natural adsorbents that contemplate all the mentioned characteristics can be promising alternatives for water treatment [65,66,73].

Adsorbents with low acquisition cost and wide availability are generally obtained from biomass originating from nature or industrial and agro-industrial processing, such as leather waste [74], rice hull [17], eggshell [18], sugar cane bagasse [75], clays [65], grape bagasse [76], among others. The production of adsorbent composites has been the focus of several studies, since it aims to improve the surface characteristics of materials from residual sources, resulting in higher capacities for removing emerging contaminants [16].

Table 3 presents recent studies that aimed the adsorption of

Table 3
Recent studies using different materials for adsorption of diclofenac.

Adsorbent material	Adsorption capacity (mg g ⁻¹)	Adsorbate		Experimental conditions			Reference
		DS	DP	pH	Equilibrium time (min)	Adsorbent dosage (mg L ⁻¹)	
Chromium (III) tanned leather waste recovered gelatin beads	36.65 ^a	X		8.15	80	2000	[74]
Composite beads from tannery wastes-derived gelatin and carbon nanotubes	20.57 ^a	X		8.15	150	2000	[16]
Thermo-plasma expanded graphite	350.00 ^b	X		1.00	10	200.00	[77]
Carbon-encapsulated iron nanoparticles	74.38 ^a	X		–	30	0.05 – 4000	[78]
Magnetic sawdust	107.52 ^a	X		6.00	–	25.00	[79]
Biochars and hydrochars derived from olive oil wastes	0.04 – 11.00 ^b	X		–	120	50,000	[69]
Rice hull biochar	14.79 – 11.84 ^a	X		2.00 and 7.00	40	25,000	[17]
<i>Anthriscus sylvestris</i> -derived activated biochar	392.94 ^a	X		6.00	1500	100.00	[80]
Ti-pillared montmorillonite clay	23.83 ^a	X		7.00	60	100.00	[81]
Clay with intercalation of Zr-pillars	55.10 ^a	X		2.00 – 4.00	≈ 100	1000	[65]
Ligand-Doped Silver Triazolate Metal-organic frameworks	91.00 – 109.00 ^a	X		10.00	90	1000	[82]
COSMOS (Colloidal Silica Medium to Obtain Safe inert: the case of incinerator fly ash)	1.70 ^a	X		–	360	15,000	[5]
Graphene oxide-based nanocomposites	18.40 – 32.37 ^a	X		4.00	120	740.00	[7]
Amino hydroxyapatite / chitosan hybrids reticulated	45.20 – 51.90 ^b	X		6.00	15 – 30	2000	[83]
Lignite activated cokes	226.00 – 790.00 ^a	X		6.50	30 – 90	25,000	[6]
MgAl/layered double hydroxide supported on <i>Syagrus coronata</i> biochar	116.52 ^a	X		5.65	60 – 360	4000	[84]
Biomass-derived carbon from eggshell	125.50 ^a	X		–	1440	–	[18]
Organophilic alkylpyridinium bentonite	91.13 ^a	X		6.00	10	2500	[71]
Aquatic plant-derived biochars	25.67 – 36.45 ^a	X		–	240	120.00	[85]
Trimethylamine functionalized clay	387.25 ^a	X		3.00	90	1500	[86]
Magnetic gelatin-activated biochar from agricultural biomass	348.90 ^a	X		6.50	600	2000	[87]
Potassium ferrate-activated porous graphitic biochar	123.45 ^a	X		6.50	1440	1000	[88]
Biochar from waste sludge/leaf	877.00 ^a	X		–	120	625.00	[89]
Metal-Loaded Carbonated Mesoporous Calcium Silicates	23.70 – 11.80 ^a	X		6.00	–	1000	[19]
Orange Peel-derived FeCl ₃ activation carbon	144.10 ^a	X		4.50	180	2000	[72]
Copper nanoparticles synthesized using <i>Tilia</i> leaves extracts	36.00 ^a		X	4.50	60	5.00	[90]
Mesoporous silica nanoparticles	333.00 ^a		X		30		
Granulated active charcoal	200.00 ^a		X	5.00	> 60	1000	[91]
Sugar cane bagasse-derived activated carbon	315.00 ^a	X		2.00	15	400.00	[75]
Magnetic amine-functionalized chitosan	469.48 ^a	X		4.50	60	400.00	[92]
Carbon nanospheres from microwave carbonized cellulose	27.30 ^a	X		5.85	–	10.00 – 100.00	[70]
Pyridine-functionalized mesoporous SBA-15 organosilicas	274.00 – 632.00 ^a	X		5.50	≈ 180	333.33	[93]
Commercial granular activated carbon	36.23 ^a	X		5.50	–	12,500	[94]
Magnetic poly (St-AMPS) nanoparticles	150.60 ^a	X		4.00	60	500.00	[95]
Activated onion skin	134.00 ^b	X		6.50	220	500.00	[96]
<i>Cyclamen persicum</i> tubers based activated carbon	22.22 ^a	X		4.00	120	14,000	[38]
Coconut shell activated carbon	103.00 ^a	X		–	≈ 480	5000	[97]
Natural Bentonite	62.50 ^a		X				
Na-Bentonite	40.00 ^a		X	2.00	150	1000	[98]
Iron-pillared Bentonite	100.00 ^a		X				
Grape bagasse	68.27 ^c	X		5.00	2200	200.00	[76]

^a Values calculated from the maximum monolayer adsorption capacity predicted by the Langmuir model.

^b Values calculated from the experimental data.

^c Values calculated from the adsorption capacity predicted by the Sips model, DS: Diclofenac sodium, and DP: Diclofenac potassium.

diclofenac using adsorbents from residual, commercial sources, or production of composites from both. The main materials produced and process parameters are presented.

Regarding the pH effect on the DCF adsorption, it is noted that a wide pH range was used (Table 3), covering the acidic and basic zones (pH = 1.00–10.00) according to each study. The DCF dissociation constant value is $pK_a = 4.15$, thus, at $pH < pK_a$, DCF molecules are positively charged, while at $pH > pK_a$, these molecules are negatively charged.

Studies that used pH close to 4.15, the greater adsorption capacity can not be explained in terms of electrostatic attraction, since in this pH range the DCF is close to neutral. Therefore, other non-electrostatic adsorption mechanisms may be responsible for high removal of DCF in highly acidic environments, which may include H-bond between the functional groups of adsorbent molecules and diclofenac [75]. Cuccarese et al. [77] assume that the closer to the neutral form of DCF at lower pH values, there is a reduction in solubility and, possibly, an increase in adsorption capacity. At more basic pH, the carboxyl groups ($-\text{COOH}$, $pK_a = 1.7\text{--}4.7$) tend to form negatively charged carboxylate groups ($-\text{COO}^-$) [6], and these adsorption sites have been reported as the main responsible for the high removal of DCF (positively charged) in this range [17]. Similar behavior has been reported in the studies by Rigueto et al. [74] and Rigueto et al. [16].

It is worth mentioning that the point of zero charge (pH_{pzc}) of each developed adsorbent material is also a parameter that will define the best pH range to be used, as well as the mechanisms that will govern the adsorption, since the surface of the adsorbent is positively charged when $pH < pH_{pzc}$, and negatively charged when $pH > pH_{pzc}$ [37].

The monitoring of the adsorption kinetics is also essential during the evaluation of new adsorbents, as it provides an estimative of the adsorption rates, the ideal amount of the adsorbent mass to be used, the time needed to end the process, in addition to suggesting possible reaction mechanisms [91,99].

The adsorption kinetics of sodium or potassium diclofenac is relatively fast, in which most studies reported that the highest removal rates were observed in the first 10 min of contact (Table 3). This fact may be associated with the rapid saturation of the materials by diclofenac in the initial minutes of contact, and later with slower rates until the balance is established.

Concerning the mathematical modeling reported in the studies analyzed, approximately 85% of the studies indicated the Pseudo-second order model with the best fit to the kinetic experimental data. In the adsorption isotherms, more than 64% of the analyzed articles state that the Langmuir model equation best described the experimental equilibrium data. The isothermal model of Langmuir assumes that every adsorption is homogeneous, it occurs in monolayers, and every active site has equivalent energy [6,67]. While the Pseudo-second order model assumes that the rate-limiting step is the surface adsorption that involves chemisorption, where the removal of a solution is due to the physical-chemical interactions between the two phases [99].

The Freundlich, Sips, Dubinin-Radushkevich, Langmuir-Freundlich, Temkin and Guggenheim-Anderson de Boer isotherm models have also been reported in some articles with good fits. Within the percentage of 15% of the works, the models of pseudo-first order, intra-particle diffusion, and Elovich were mentioned with good fits.

The adsorption capacities of the materials presented in Table 3 varied from 0.04 to 877.00 mg g^{-1} . Commercial materials did not always provide the highest adsorption capacities, which were mainly obtained by materials manufactured from residual sources with or without functionalization, such as *Anthriscus sylvestris*-derived activated biochar [80], lignite activated cokes [6], trimethylamine functionalized clay [86], magnetic gelatin-activated biochar from agricultural biomass [87], biochar from waste sludge/leaf [89], sugarcane bagasse-derived activated carbon [75], magnetic amine-functionalized chitosan [92], among others.

Despite the variety of promising materials that can be used as DCF adsorbents, a major challenge is the practical application of these

materials is the life cycle of the adsorbent material, which includes considerations about the possibility of regeneration and separation of the material, in addition to economic aspects [66].

In order to overcome these limitations, in the current works, the regeneration/reuse tests are being included to justify the applicability of the material. Chauhan et al. [81] verified the reuse efficiency of 62.13% of Ti-pillared montmorillonite clay during three DCF adsorption/desorption cycles, and the best eluents for promoting DCF desorption were HCl, HNO_3 , and H_3PO_4 solutions. El Naga et al. [75] reported removal efficiency of 92.4% of DCF after 5 cycles using sugarcane bagasse-derived activated carbon and acetone as eluent, with no changes in the structural integrity of the adsorbent after use.

Cuccarese et al. [77] demonstrated after 4 cycles that desorption of DCF in Thermo-plasma expanded graphite was more efficient ($\approx 100\%$) employing thermal recovery compared to recovery by solvent washing, since the high solubility of DCF in basic solutions was not enough to desorb DCF, and just by heating all the material, the DCF adsorbed on it is released and all the sites available for adsorption interact with other molecules in a new cycle.

For the separation purposes of the adsorbent material after use, transition metals (Fe, Ni, Co etc) are being added to the adsorbent composites, allowing the separation of the adsorbent from aqueous solutions with the aid of an external magnetic field [87,95]. Tam et al. [87] reported that iron oxide (Fe_3O_4) loaded on the surface of gelatin-activated biochar from agricultural biomass was effective in creating the magnetism, and in this way, the biochar becomes easy to be isolated from the aqueous solution using a magnetic separator.

Finally, even with promising application trends, the continuity of studies that promote the development of adsorbent materials for pharmaceutical removal, such as diclofenac, is essential. We emphasize that in addition to batch tests, tests on fixed bed columns are also necessary, since they allow the simulation of the adsorbent application in continuous systems [100], as well as using real effluents.

4.2. Membrane separation

After World War II, membrane separation technology has begun to play an essential role in the pharmaceutical industry, medical applications, and the field of microbiology and, due to its flexibility, a variety of processes have been developed to mitigate several industrial challenges [101], showing promising results, such as high separation efficiency in wastewater treatment, water quality and scalability [102].

Membrane is a barrier used in the transport of substances under a potential gradient (mechanical, chemical or electrical), acting as a physical barrier that divides the flow of feed into permeate and concentrate, with the most popular modules being in the form of plate, tubular, spiral and hollow fiber [101].

Membranes can be produced from different materials (polymeric, ceramic or mixed matrix compounds), shapes and morphologies, porous or non-porous, charged or neutral, and solid or liquid [101]. The material that composes it must be chosen considering the use of raw materials that limit production costs and, at the same time, achieve the desired corrosion resistance and mechanical, structural and filtration properties in the final membrane support [102]. The method of preparation and the composition of the membrane determines the properties and effects [103].

The membrane processes are classified into reverse osmosis (RO), nanofiltration (NF), ultrafiltration (UF), and microfiltration (MF) [101], according to the pore size of the membrane (Fig. 3).

The operating conditions of the membrane are essential to obtain efficiency in the removal of pharmaceuticals from aqueous matrices. Reverse osmosis membranes, for example, have great rejection efficiencies, but present higher operating costs compared to the others due to the need for high operating pressure (to overcome the lower permeability); however, they have advantages in the removal efficiency by having a smaller pore size compared to other membrane separation

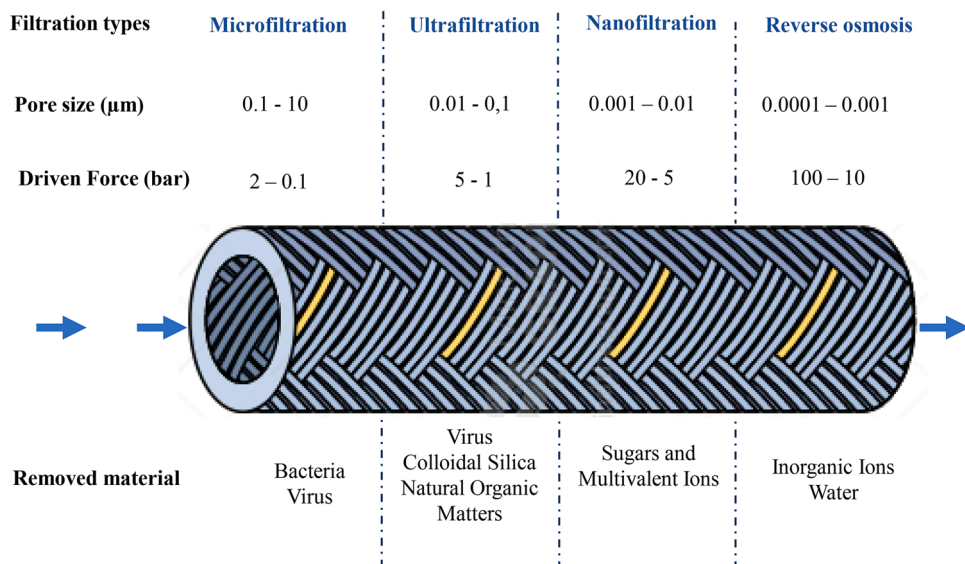


Fig. 3. Classification of membranes according to pore size, driven force and removed materials.

processes [23]. Thus, it is necessary to evaluate the entire process involved (purpose/application of membrane and contaminant to be removed), linking the feasibility of its use.

One of the main advantages of membranes is the removal of contaminants in low concentrations and/or dissolved in colloidal form. In this context, many studies have been conducted using membrane separation to remove pharmaceuticals, such as caffeine [21,104], ibuprofen [105], diclofenac [106,107], among others. Table 4 presents some studies that used membranes to remove diclofenac from aqueous media, indicating whether there was the integration of other processes, operational conditions tested, and the removal efficiencies obtained.

Nanofiltration and reverse osmosis are considered more effective methods for the removal of pharmaceuticals since they have smaller pores, reaching removal > 90% when used individually [22,23,105,117] or combined [20,21,115,116,118,119]. When using microfiltration and ultrafiltration, it was necessary to employ other pre- or post-treatment techniques to achieve higher removal rates, since the molecular sizes of pharmaceuticals are generally smaller than the pores of the respective membranes [9,104,106–114].

The rejection of diclofenac is complex and influenced by several

factors, such as physicochemical properties of the solute, membrane properties, chemistry of the solution, and operational conditions [120]. In this sense, the removal percentages shown in Table 4 are varied, related to different sources of aqueous media tested and the approaches combining membrane separation with other techniques, such as adsorption, electric field, coagulation, among others.

Despite this complexity, exclusion by size is always considered the most important rejection mechanism. Licon et al. [23] assessed the potential of nanofiltration and reverse osmosis to remove diclofenac from water. As expected, the reverse osmosis process was more efficient (98% removal) compared to nanofiltration ($\approx$ 90% removal), in which removal efficiencies are mainly attributed to the membrane pore size.

However, some studies go beyond this consideration, relating the contribution of electrostatic interactions and steric effects to the rejection of diclofenac, attributed to an anionic form of the diclofenac molecule at pH 5 and 7, which can significantly influence the separation efficiency [33]. Molecular rejection is carried out with the help of Donnan exclusion when positively or negatively charged molecular species present in the solvent interact with charges present on the membrane surface or matrix through the construction of an interface.

Table 4

Studies on membrane separation processes for diclofenac removal.

Filtration processes	Integrated processes	Operating conditions				Diclofenac removal (%)	Reference
		Pore size (μm)	Flow rate (m ³ h ⁻¹)	Area (m ²)	pH		
Microfiltration	TiO ₂ nanoparticles Photocatalysts	0.22	–	0.00159	7	68	[108]
	Modified Ludzack – Ettinger tanks	0.45	–	–	6 and 8	44.4	[104]
	Heterogeneous Fenton oxidation	–	–	0.0085	3 and 7.5	65	[109]
	Electric field	0.1	–	–	3–8	91.6	[110]
	Adsorption Coagulation	0.1	–	0.75	11	> 90	[106]
Ultrafiltration	Adsorption Ultrasound irradiation	–	–	0.0006	–	100	[111]
	Coagulation and adsorption	–	1.36	0.024	7.5	100	[112]
	Filtration and addition of surfactants	–	–	0.0028	–	100	[113]
	Electric field	0.04	–	0.047	7 and 8	80	[114]
	Adsorption and ultrasound irradiation	0.1	–	0.0006	–	±98	[107]
Nanofiltration	Ultrafiltration	–	–	–	6.11–8.5	> 95	[115]
	–	0.0004	–	0.0042	3–10	94.1	[22]
	Coagulation and adsorption	–	–	0.014	3 and 8	> 95	[21]
	Glass filter (0.7 μm) and ultrasonicated	–	–	0.024	1.5–10.5	97	[20]
	–	–	–	0.0064	2–12	99.7	[105]
Reverse osmosis	Activated sludge and ultrafiltration	–	–	1295	–	95	[116]
	–	–	–	0.004	7	90	[117]
	Nanofiltration	–	–	0.8	7	> 90	[118]
	–	–	0.04	0.028	2–11	98	[23]
	Photocatalytic (N and TiO ₂) Visible irradiation	0.05	–	0.0148	6.5	95	[119]

This relationship was verified by Tran and Gin [104] who studied the removal of diclofenac through microfiltration and modified Ludzack-Ettinger tanks, attributing the low removal (maximum of 44.4%) to the absence of electron donor groups and/or predominance of strong electron withdrawal groups in diclofenac molecules, as this contains only strong electron withdrawal groups (CO-NH_2 , $\text{Cl-CH}_2\text{-COOH}$), and the absence of electron donor groups results in low removal.

One of the main challenges for applicability of membrane separation is the fouling phenomenon, which occurs due to the polarization of the concentration with the accumulation of solute on the membrane surface or in the pores, reducing the performance of the membrane and limiting its life cycle [102]. The life cycle assessment is carried out periodically through input and output data for systemic tracking, classified as material, energy and emission flow, contributing to necessary adjustments throughout the process [102,103]. To reduce the propensity for fouling, it is necessary to apply appropriate pre-treatments to each sample, aiming to reduce the costs related to cleaning the membrane, as well as increasing its useful life. For this reason, membrane separation is utilized as a tertiary treatment for wastewater purification [120].

Studies have aimed at the use of membrane separation and integrated membrane system to remove diclofenac, showing significant removal results (over 90%), usually attributed to specific operational conditions [103], as the size of the membrane pore, where the high removal of diclofenac occurred due to its molecular size [23,117], electrostatic effect and steric impediment [22], and pH [105].

The integrated process approach can further increase the removal efficiency compared to conventional membranes [121]. Besha et al. [122] addressed the removal of diclofenac by the individual membrane bioreactor technique and integrated it with other membrane separation processes (reverse osmosis and nanofiltration). The removal results showed relevant differences, ranging from 15% to 87% by membrane bioreactor, 87–97% by membrane bioreactor and nanofiltration, and 88–95.9% by membrane bioreactor and reverse osmosis.

As for integrated processes, studies using membrane bioreactors have become well established for wastewater treatment, with a tendency to represent an \$ 8.27 billion market by 2025. The use of integrated processes is based on conventional activated sludge treatment and membrane separation, which replaces the conventional sedimentation tank [123]. Another approach that has been widely evaluated is the combination of reverse osmosis and a direct osmosis membrane system, usually applying direct osmosis to pump wastewater, while reverse osmosis separates and recycles the water withdrawn, showing promising trends in the removal of contaminants, in addition to minimizing fouling on membranes [121].

Besides, there are prospects that membrane separation can be performed directly at urban wastewater treatment plants, which are the main barriers to prevent emerging contaminants, such as diclofenac, for being detected in surface and drinking water. Thus, even with challenges to be overcome, the membrane separation process can be listed as a viable local implantation and large-scale technique [101].

4.3. Advanced oxidation processes

Advanced oxidation processes (AOPs) can also be used as tertiary techniques to remove emerging contaminants. These processes are chemical, electrochemical or photochemical methods, which form oxygen free radicals in situ, mainly hydroxyl radicals ($\cdot\text{OH}$) with high oxidizing power, which promote the degradation or combustion of these organic compounds, through hydroxylation or dehydroxylation reactions until the formation of carbon dioxide, water, inorganic ions and other molecules of lower molecular weight [9,124,125].

On the other hand, AOPs can form intermediates due to the complexity of the compounds in the media, which can lead to products with greater toxicity, requiring the use of combined techniques to improve the efficiency of contaminant removal. For this purpose, it is

possible to use chemicals products or membrane separation processes, and also combinations of oxidation processes, such as photocatalytic ozonation, photoelectrocatalysis, among others [9124].

AOPs can be divided into different techniques according to different classifications. In this review, they were divided according to the generation source of oxidizing species, which are based on ozone, hydrogen peroxide, photolysis, electrochemical oxidation, sonochemical oxidation and heterogeneous photocatalysis, with a possible source of each method illustrated in Fig. 4 [125].

Photolysis uses energy in the form of photons through UV radiation, which inserts this energy into the molecules present in the medium, causing their specific electrons to reach an excited state, which forms free radicals and initiating chain reactions until obtaining a final product [9,125]. These radicals are formed by homolysis of weak bonds or the transfer of electrons to oxygen molecules. The advantages are the absence of chemical products, low maintenance and operation costs; however, its effectiveness can be compromised due to the formation of organic molecules that can cause turbidity. Thus, this technique is commonly applied with other oxidants to enhance the production of radicals [125].

Another method of OAPs is ozonation (O_3), which removes organic contaminants by direct reaction with dissolved O_3 or indirectly through the formation of OH. In direct oxidation, ozone selectively reacts with functional groups rich in electrons (amines, activated aromatic rings and others); on the other hand, in indirect oxidation, it decomposes, in a basic medium, with the hydroxyl radicals OH, which interact with most molecules [124,126]. Ozonation may have its potential to form hydroxyl radicals increased due to combinations with other oxidizing agents, such as O_3/UV , $\text{O}_3/\text{H}_2\text{O}_2$ e $\text{O}_3/\text{UV}/\text{H}_2\text{O}_2$. Ozonation is favored in a basic medium and in the addition of hydrogen peroxide, due to the higher density of OH ions in the medium and the chain reaction with H_2O_2 [9]. UV can be used to cleave dissolved O_3 , forming O_2 that reacts with water molecules, generating H_2O_2 in excited form, which decomposes into 2OH [126]. Furthermore, it is possible to combine the 3 oxidants ($\text{O}_3/\text{UV}/\text{H}_2\text{O}_2$), in which direct photolysis, the action of O_3 , alone, UV photolysis of O_3 , the reaction of O_3 with H_2O_2 and UV photolysis of H_2O_2 [125].

When used alone, AOPs based on hydrogen peroxide do not show high efficiency, since they do not have good oxidizing properties. Therefore, they must be applied in combination with other compounds, such as UV radiation, Fe^{2+} (Fenton) and Fe^{3+} (Fenton-like) ions, being possible the combination of three techniques such as $\text{UV}/\text{H}_2\text{O}_2/\text{Fe}^{2+}$ or Fe^{3+} (Photo-Fenton) [125]. The reaction between H_2O_2 and Fe^{2+} occurs in an acid medium ($\text{pH} = 2.8\text{--}3.0$), making Fe^{2+} from Fe^{3+} , OH radicals and intermediates. Similarly, the reaction between Fe^{3+} and H_2O_2 is carried out with the same purpose, but with the difference that the Fe^{3+} ions are reduced to Fe^{2+} to form the OH radicals, the main difference in its use is in relation to the price of the iron ions, with Fe^{+3} being the cheapest [125,127].

Another method based on H_2O_2 is the Photo-Fenton, which uses iron ions (photocatalysts), hydrogen peroxide (oxidizing agent) and UV radiation, for the greater formation of OH radicals [125]. In this process, photoreduction of Fe^{3+} into Fe^{2+} occurs, which reacts with H_2O_2 to form OH radicals; simultaneously, the reduction reaction of Fe^{3+} forms ferric aqua-complexes, which are also reduced by UV radiation, generating OH groups [124]. Techniques that use hydrogen peroxide have advantages that consist of easy operation and efficiency in the degradation of contaminants, however, they have limitations, such as the presence of iron in the sludge, which causes secondary pollution, in addition to requiring acidic conditions for the reaction, which raises costs due to pH correction [127].

The sonochemical processes carry out sonification of the effluent through ultrasound in the range of 20–500 kHz. Microbubbles are formed through the phenomenon of ultrasonic cavitation and, when reaching a critical resonance size, they implode generating high temperatures ($>5000\text{ K}$) and pressures ($>1000\text{ bar}$) inside the bubble for a

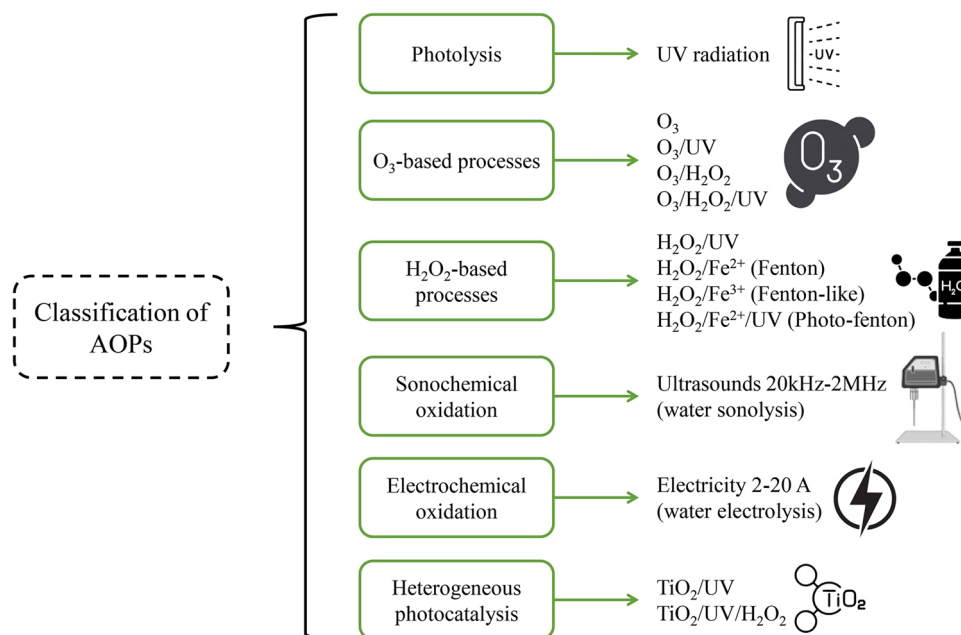


Fig. 4. Classification of advanced oxidation processes (AOPs) according to the source of .OH radicals.

short period of time (milliseconds) [126]. Thus, two mechanisms are involved in the degradation of contaminants: the thermal cleavage of molecules into H and OH radicals and their oxidation by OH radicals [125]. Sonochemical oxidation has a low electrical efficiency compared to other techniques, so it can be used in a combination with other technologies, such as UV, O₃, H₂O₂ and catalysts (TiO₂), to increase this efficiency [126].

Electrochemical oxidation is the oxidation of organic molecules by electric currents to form non-toxic compounds [127]. Oxidation can be direct or indirect: the first one occurs by the transfer of electrons between the electrode (anode) and the contaminant, and the second by the formation of oxidation intermediates (OH) due to the reaction of water anions with the anode, being these radicals adsorbed on the electrode surface [9].

The electrode material influences the efficiency of the technique, since they can generate partial oxidation of contaminants (active electrodes) or complete formation of CO₂ (non-active electrodes) [9,128]. Some examples of active electrodes are from RuO₂, Pt and doped Ti electrodes and inactive ones consist of boron-doped diamond, Pb and Sn oxides [128]. This process can also use chlorine ions that help in the formation of oxidative radicals, and when in the form of sodium chloride, they generate lower consumption due to the increased conductivity of the solution [128].

Heterogeneous catalysis consists of a semiconductor material (solid phase) and contaminants (liquid phase), in which reactions occur on the catalyst surface [125]. In this case, the semiconductor is irradiated by UV light, absorbing the energy in the form of photons, which, upon reaching the surface, excite the electrons to the conduction band, leaving a positively charged hole in the valence band [126]. These electrons cause a reduction of the oxygen present in the water and the spaces left by them, oxidation of the water constituents occur, forming OH radicals [125,126]. In this method, hydrogen peroxide can be inserted to increase the photodegradation rate due to its dissociation into two OH radicals, increasing their density in the medium [125].

The most used catalyst is titanium dioxide (TiO₂), which has advantages such as low cost, non-toxicity, chemical stability, reuse potential and contaminant removal efficiency [125,126]. The main difficulties of application on a large scale are related to the uniform distribution of radiation on the surface of the catalyst and the need for a treatment to purify and separate the catalyst from the water suspension

[125].

AOPs and their different techniques and combinations can be used to remove several emerging contaminants, such as pharmaceuticals, which are the most varied, such as azithromycin, carbamazepine, sulfamethoxazole, tetracycline, oxolinic acid, diclofenac and others [24,129]. Table 5 presents some AOPs used in the removal of diclofenac from synthetic and real aqueous matrices.

As shown in Table 5, different approaches of AOPs have been studied to degrade diclofenac in aqueous medium. Efficiencies vary according to techniques, diclofenac concentration, types and doses of catalysts, pH, treatment time, among other operating conditions. Diclofenac has been reported as one of the top 5 pharmaceutical contaminants removed in AOPs [9]. Photo-Fenton and ozone-based processes are the main techniques studied by Ibero-American countries [124], while Fenton-based processes, photocatalysis, and ozonation are the most used technologies in the European context [145].

Another important aspect to be considered is whether the medium to be treated is a real sample or synthetic effluent. Real field samples may contain certain characteristics that influence their treatment. In the case of electro-oxidation processes, real effluents can have other radical electrolyte species, which can affect the treatment efficiency [140]. Serna-Galvis et al. [146] report an increase in diclofenac concentration with longer ultrasound treatment. These authors attribute this behavior to the fact that some contaminants are retained in the suspended solids of the effluent due to their high interaction with organic matter and, when using the ultrasound method, they may be released from the solid matrix to aqueous.

The diclofenac presence in environmental compartments presents risks to different organisms, and this risk varies according to its concentration [147]. Generally, the effects of diclofenac are measured using the risk quotient (RQ), when the $RQ < 1$, there is no potential ecotoxicological risk [148]. After the treatment performed by Martínez-Pachón et al. [129], the RQ value for diclofenac ranged from medium to low.

Although they have high efficiency for diclofenac degradation, AOPs can generate intermediate products that are more toxic and/or resistant than the original compound [9]. This aspect can be considered one of the main bottlenecks of AOPs. Another discussion in the scientific community is about the economic and environmental viability of different AOP techniques, in which some researchers consider AOPs as cost-effective techniques for wastewater treatment. On the other hand, it is

Table 5

Diclofenac treatment by advanced oxidation processes in different matrices.

Technology	Nature of sample	Removal efficiency (%)	Experimental conditions	Reference
Photo-electro-Fenton	MWTP effluent	≈ 85	Current density: 6.92 mA cm ⁻² ; pH: 3.0; Temperature: 20 °C; Agitation: 400 rpm (constant); Batch cell; Cathode (carbon-felt) and a stable anode (DSA, Ti/IrO ₂ doped with SnO ₂)	[129]
UV radiation	Synthetic	88.5 ± 3.8	Contact time: 15 min; UV lamp: 8 watt low-pressure	[24]
Sodium hypochlorite		100	Sodium Hypochlorite: 4 mg L ⁻¹ ; Contact time: 120 min	
UV/HOCl		100	Sodium Hypochlorite: 5 mg L ⁻¹ ; Contact time: 15 min	
UV/H ₂ O ₂		100	H ₂ O ₂ : 5 mg L ⁻¹ ; Contact time: 5 min	
Heterogeneous Fenton	Synthetic	99.65	H ₂ O ₂ : 0.05 mol L ⁻¹ ; Pd loading: 5%; pH: 3; [DCF]: 10 mg L ⁻¹ ; time: 40 min	[25]
Photocatalysis	Synthetic	93.65	Best ratio of Bi ₂ S ₃ /BiOBr/BC: 1% (Bi ₂ S ₃) and 10% (BC); Time of visible LED light: 40 min; Photocatalyst dosage: 0.6 g L ⁻¹ ; pH: 5.0.	[130]
Photocatalysis	Synthetic	98.7	Batch reactor; Mercury lamp: 80 W; [DCF]:15 ppm; pH: 10.7; Photocatalyst: 30 mg; Time: 3 h	[26]
Heterogeneous Fenton	Synthetic	97	Reaction temperature: 25 °C; H ₂ O ₂ concentration: 10 mM; pH: 3; Catalyst concentration: 2 g L ⁻¹ ; time: 15 min	[131]
Photocatalysis	Synthetic	98	[DCF]: 0.15 mg L ⁻¹ ; Catalyst dose: 0.988 g L ⁻¹ ; pH: 5.5.	[132]
Three-dimensional electro-Fenton-persulfate	Synthetic	96.3	[DCF]: 0.05 mg L ⁻¹ ; Persulfate: 1.50 mM; Fe0: 3.00 mM; Anode: titanium mesh; Cathode: carbon fiber; pH: 3, Tank voltage: 8.00 V	[27]
Heterogeneous electro-Fenton	Synthetic	96	[DCF]: 140 mg L ⁻¹ ; pH: 6; Na ₂ SO ₄ : 0.01 M; Air: 1 L min ⁻¹ ; Intensity: 300 mA; Catalyst dose: 20 g L ⁻¹ .	[133]
Biological-Sonochemistry	Hospital wastewater	67.42	90 min of treatment; Frequency: 375 kHz; P: 88 W L ⁻¹ ; pH: 7.9; [NOR] 40 μmol L ⁻¹ ; [Fe ²⁺]: 90 μmol L ⁻¹ ; UVC: 4 W	[134]
Biological-Sono-photo-Fenton	Synthetic	100	Photocatalyst: 0.1 g Fe ³⁺ -TiO _{2-x} N _x synthesized by microwave; Visible light lamp: 23 W; Time: 300 min; pH: 3.5	[135]
Photocatalysis		72.3	TiO ₂ doped with Boron; Reaction time: 120 min; 100 mgcatalyst dm ⁻³ ; 15 mgcontaminant dm ⁻³ ; 5 wt% dopant.	
Photocatalysis	Synthetic	≈ 35		[136]
O ₃	Synthetic	≈ 60		[137]
TiO ₂ + O ₃		86	3 min of treatment; [DCF]: 10 mg L ⁻¹ ; Aeration intensity: L min ⁻¹ ; Power of reactor: 50 W,	
UV + O ₃		76	Power of UVC lamp: 40 W; Time of pretreatment in reactor: 2 min	
TiO ₂ + UV + O ₃		100		
Gamma-irradiation	Synthetic	≈ 95	[DCF]: 4.5 mg L ⁻¹ ; Gamma radiations: 1015 Gy; Dose rate: 290 Gy h ⁻¹	[138]
Heterogeneous photo-Fenton	Synthetic	98.9	UV-A/FeZSM5/H ₂ O ₂ process; optimal condition: pH: 4; [H ₂ O ₂]: 50 mM: [Fe]FeZSM5: 2.0 mM; 2 h of treatment	[139]
Electro-oxidation	Hospital wastewater	≈ 95	[DCF]: 62.5 μg L ⁻¹ ; Reaction time: 40 min; Phosphate buffer electrolyte with Pt electrode	[140]
Acoustic cavitation + electrochemical oxidation	Synthetic	93	Reaction time: 5 min; [DCF]: 0.157 μmol; Na ₂ SO ₄ : 30 mmol L ⁻¹ ; Frequency: 850 kHz; Reaction volume: 500 mL	[141]
Ozonation	WWTP effluents	> 90	5–7 mg L ⁻¹ ozone; Reaction time: 10 min; Bench-scale batch	[142]
Electro-oxidation + electrocoagulation	Synthetic	84	Stainless steel anode; Specific charge: 4200 Coulombs L ⁻¹ ; pH: 10.9	[143]
Sonochemical oxidation	Synthetic	100	Reactor: multi-frequency plate-type; Piezo-electric transducer: adjusted to emit at 861 kHz; 90 min sonication; DCF concentration: 5 mg L ⁻¹ ; pH: near neutral	[144]

Legend: MWTP: municipal wastewater treatment plant; WWTP: wastewater treatment plant; [DCF]: diclofenac concentration.

necessary to develop studies, such as life cycle assessments, to determine the economic, environmental, and social viability of using different approaches of AOPs to treat aqueous matrices [9124].

Integrated technologies have been proposed as a way to increase the efficiency and feasibility of techniques for treating emerging contaminants, including diclofenac. Studies have shown that the integration of different AOPs with membrane separation, biological processes, or even with more than one AOP, can result in more viable technologies since there is more production or better performance of hydroxyl radicals for the degradation of contaminants [128,134].

Regarding the life cycle (LCA) of the AOP's, it is taking into account the formation of by-products from the partial purification of contaminants, remaining oxidants, intensive use of chemicals, energy consumption, in addition to CO₂ and other gases emissions and solid waste generated [149]. In the study by Rodríguez et al. [149], the LCA of conventional Fenton process and heterogenous Fenton-like applied to the treatment of pharmaceutical wastewater were evaluated. These authors report that homogeneous oxidation has as a disadvantage the formation of a toxic sludge containing metals (0.63 kg of waste per process) and higher energy demand due to process temperature, sludge drying and agitation. In the heterogeneous process, the hazardous waste formed is catalyst loaded with metal ions (0.0145 g of waste per process), in which it uses a lower temperature, though, with greater use of chemical reagents, with hydrogen peroxide being the largest portion.

Advanced oxidation processes in the heterogeneous phase have advantages over homogeneous ones, due to the lower amount of toxic waste generated and mild operating conditions. However, some points need to be more explored, such as the issue of intermediate toxic contaminants generated by AOPs and the release of toxic gaseous emissions, in addition, the expansion of researches on an industrial scale and not just laboratory for the analysis of LCA's for different combinations of AOPs is essential for practical application purposes [149].

5. How diclofenac treatment technologies can contribute to achieving sustainable development goals?

In 2015, the United Nations approved the 2030 Agenda for Sustainable Development, which comprises 17 Sustainable Development Goals (SDGs) and 169 targets, aiming to create efforts to solve environmental, economic, and social challenges [150]. Several researchers have been relating their studies with the SDGs given the importance of this theme for all segments of society [151].

As shown in this work, varying diclofenac concentrations are reported in aquatic environments around the world, which represent risks to human health and affect the development of different organisms, such as vultures, aquatic animals, soil invertebrates, plants, and mammals [147]. Therefore, the presence of diclofenac in different environmental compartments can negatively impact the achievement of some SDGs,

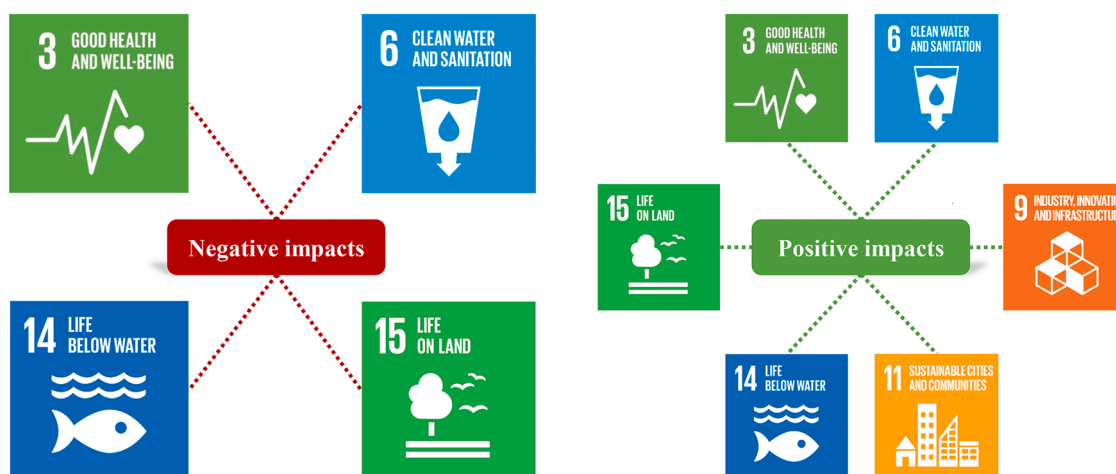


Fig. 5. Negative impacts related to the presence of diclofenac in different environmental compartments and the positive impacts of applying diclofenac removal techniques, according to the SDGs.

such as Good Health and Well-Being (SDG 3), Clean Water and Sanitation (SDG 6), Life Below Water (SDG 14), and Life on Land (SDG 15) (Fig. 5).

In this sense, it is necessary to improve technologies capable of reducing the concentration of emerging contaminants in aquatic matrices, including diclofenac, through different treatment approaches [100]. The more applicable to the different realities of the countries these technologies are, the more they will be contributing to achieve the SDGs. The application of sustainability concepts in the development of technologies for the diclofenac treatment can positively impact several SDGs, such as Good Health and Well-Being (SDG 3), Clean Water and Sanitation (SDG 6), Industry, Innovation and Infrastructure (SDG 9), Sustainable Cities and Communities (SDG 11), Life Below Water (SDG 14), and Life on Land (SDG 15) (Fig. 5).

The SDGs positively impacted by diclofenac treatment technologies (Fig. 5) were selected considering that reducing the concentration of this contaminant results in better health conditions for the population (SDG 3) by providing drinking water with more adequate treatment (SDG 6). Moreover, when applied to effluent treatment, these technologies reduce the diclofenac concentration in surface and groundwater. Although it is not the focus of this work, it is also possible to relate the SDG 15 since many crops demand water for irrigation, which can increase the presence of diclofenac in the soil. Thus, the appropriate treatment of diclofenac can reduce the impacts of this contaminant on terrestrial compartments (SDG 15). Finally, the positive impacts on SDG 9 and SDG 11 result from the current need to seek innovative solutions for the development of more sustainable treatment technologies that, above all, apply to the reality of each country.

6. Conclusions and perspectives

This paper addressed a review of the main tertiary techniques to diclofenac removal from synthetic aqueous media and real effluents. The techniques reviewed were adsorption, membrane separation and advanced oxidation processes, presenting efficiencies of DCF removal and mechanisms, besides their advantages and limitations. Moreover, it was presented the global occurrence of this emerging contaminant, showing that DCF is detected in concentrations ranging from ng.L^{-1} to mg.L^{-1} .

The investigated techniques showed promising results for the diclofenac removal (up to 100%) from aqueous matrices. On the other hand, some challenges need to be overcome, which in general involve the generation of more toxic compounds than diclofenac, high cost of chemical reagents and equipment, generation of sludge or concentrate medium containing highly diclofenac concentration, need for recovery

and reuse of adsorbents at the end of the process, incrustation of membranes, among others. In this sense, researchers, policy makers, and stakeholders must focus on developing more sustainable technologies since a lot of factors are neglected, such as cost analysis, life-cycle assessment, and disposal of sludge or concentrate media, and therefore more studies are needed to overcome these gaps.

The implementation of integrated technologies, such as hybrid systems, is a promising alternative to increasing the effectiveness and viability of the real application of these techniques. Finally, when sustainability is applied for the development of technologies to emerging contaminants treatment, several Sustainable Development Goals can be positively impacted, especially those related to human health, the maintenance of aquatic and terrestrial ecosystems, industry innovation and sustainable cities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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